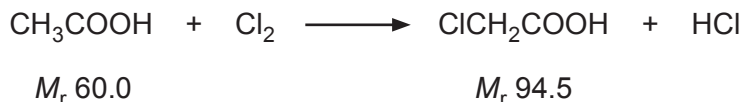


SECTION B

Answer **all** questions in the spaces provided.

8. (a) Chloroethanoic acid is produced from ethanoic acid by passing chlorine gas into it under suitable conditions.



- (i) This reaction involves radicals. In the first stage chlorine radicals are formed. They then attack the ethanoic acid molecule producing a molecule of hydrogen chloride and a new radical.

Suggest the formula of the new radical formed from the acid during this reaction.

[1]

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- (ii) I. In an experiment 89.0 g of ethanoic acid reacted with chlorine.

Calculate the increase in mass that occurs if chloroethanoic acid is the only organic product. [2]

Increase in mass = g

- II. Chloroethanoic acid reacts with ammonia producing aminoethanoic acid.



In the experiment described in I, 89.0 g of ethanoic acid then gave 49.2 g of aminoethanoic acid (M_r 75.0).

Calculate the percentage yield of aminoethanoic acid. Give your answer to an **appropriate** number of significant figures. [2]

Percentage yield = %



- III. Aminoethanoic acid is isolated from the solution of the products by adding methanol to the liquid. The acid is precipitated as a white solid.

Suggest why aminoethanoic acid is insoluble in methanol. [2]

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- (b) Explain why an aqueous solution of aminoethanoic acid does not affect the plane of plane polarised light. [1]

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- (c) Draw the structure of the dipeptide formed from two molecules of aminoethanoic acid. [1]



- (d) Aminoethanoic acid reacts with nitric(III) acid to produce nitrogen gas.



Calculate the maximum volume of nitrogen gas produced at 100 °C and 98 kPa pressure when 0.300 mol of aminoethanoic acid reacts in this way. Give your answer in dm³. [3]

Volume produced = dm³

- (e) Hydroxyethanoic acid, CH₂(OH)COOH, can also be prepared by reacting chloroethanoic acid with aqueous sodium hydroxide.

Explain why the reaction mixture needs to be acidified to produce hydroxyethanoic acid. [1]

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